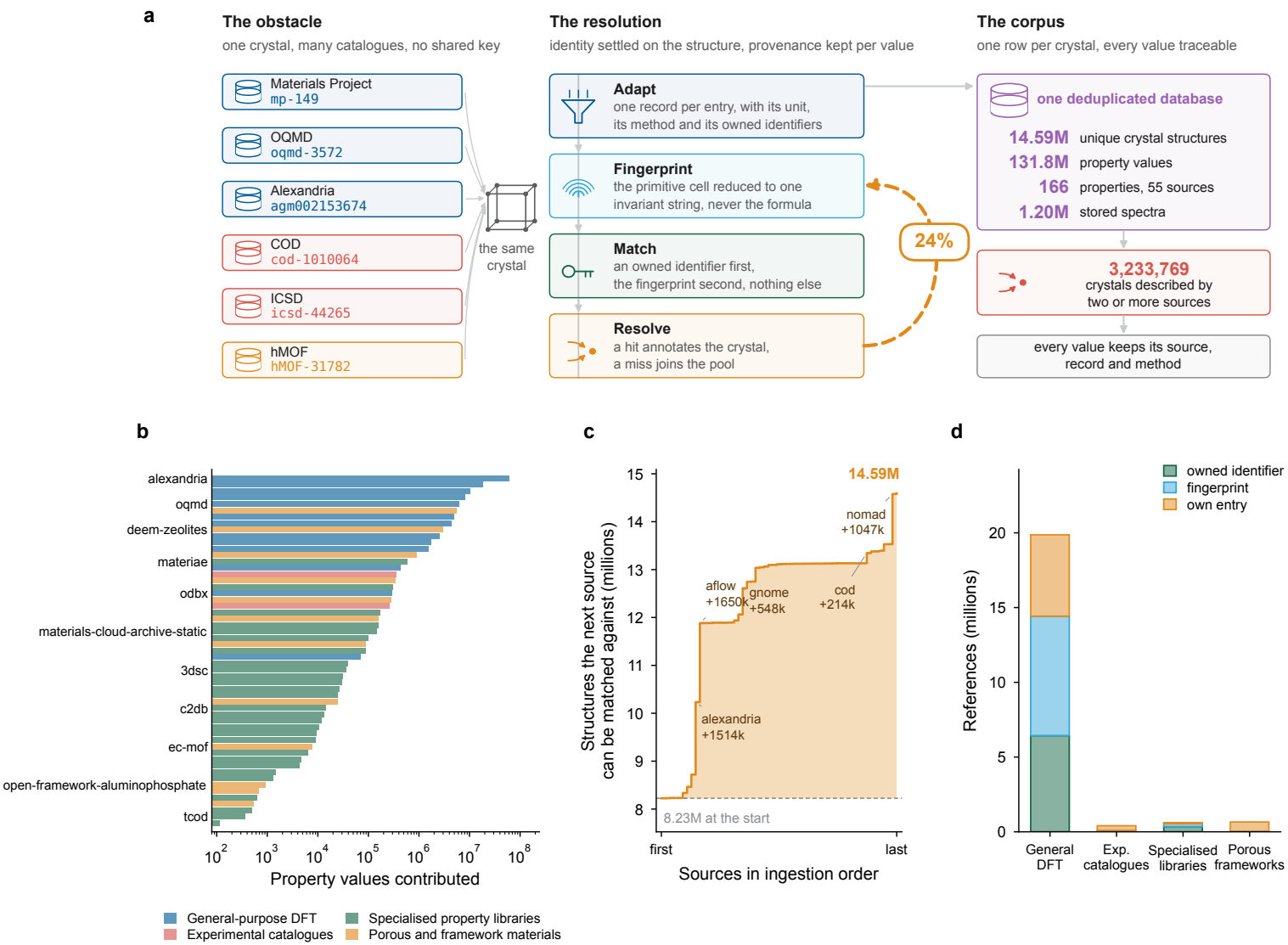

a
The obstacle
one crystal, many catalogues, no shared key
Materials Project
mp-149
OQMD
oqmd-3572
Alexandria
agm002153674
COD
cod-1010064
ICSD
icsd-44265
hMOF
hMOF-31782
the same crystal
The resolution
identity settled on the structure, provenance kept per value
Adapt
one record per entry, with its unit, its method and its owned identifiers
Fingerprint
the primitive cell reduced to one invariant string, never the formula
Match
an owned identifier first, the fingerprint second, nothing else
Resolve
a hit annotates the crystal, a miss joins the pool
24%
The corpus
one row per crystal, every value traceable
one deduplicated database
14.59M unique crystal structures
131.8M property values
166 properties, 55 sources
1.20M stored spectra
3,233,769
crystals described by two or more sources
every value keeps its source, record and method
b
alexandria
oqmd
deem-zeolites
materiae
odbx
materials-cloud-archive-static
3dsc
c2db
ec-mof
open-framework-aluminophosphate
tcod
Property values contributed
General-purpose DFT
Experimental catalogues
Specialised property libraries
Porous and framework materials
c
14.59M
nomad +1047k
aflow +1650k
gnome +548k
cod +214k
alexandria +1514k
8.23M at the start
Structures the next source can be matched against (millions)
first
last
Sources in ingestion order
d
owned identifier
fingerprint
own entry
References (millions)
General DFT
Exp. catalogues
Specialised libraries
Porous frameworks